

Stephan Zwicknagl

Software Engineer

github.com/stephanzwicknagl | [linkedin.com/in/stephan-zwicknagl](https://www.linkedin.com/in/stephan-zwicknagl) | U.S. Green Card Holder

EDUCATION

Master of Science in Computational Science, *University of Potsdam, Potsdam, Germany* Jan 2024

Thesis: Full stack development of a Flask-based dashboard for explanations of AI decisions using Python and ReactJS. Conceived intuitive explanations of logic programs. <https://github.com/potassco/viasp>

Bachelor of Science in Physics, *University of Regensburg, Regensburg, Germany* Sep 2020

Thesis: Adapted a computational model to investigate high harmonic generation in several semiconductor materials.

SKILLS

5+ years: Python, Statistical Data Analysis (Numpy, Pandas, Excel), Git, JavaScript, Docker, HTML

1-5 years: Machine Learning (Scikit-learn, Keras, TensorFlow), SQL Database, Github Actions, CD/CI

<1 year: TypeScript, React Native, Tailwind CSS, Google Cloud Platform

Languages: English (fluent), German (native)

EXPERIENCE

Software Engineer

Nov 2023 - Apr 2025

Potassco GmbH, Germany

Arlington, VA (Remote)

- Led development of open source visualization tool and accelerated its integration in workflow of 14 engineers.
- Conceived of a novel approach to tracing AI decision making based on internal requirements and engineered a React+Flask app.
- Implemented a CI/CD pipeline with automated testing and package publishing to PyPI and Docker using GitHub Actions.
- Delivered technical presentations to a team of engineers and student groups, demonstrating the tool and providing hands-on guidance, significantly improving adoption rates.
- Authored detailed technical documentation and user tutorials to support effective tool usage and provide clear explanations of system components.

Teaching Assistant

Nov 2022 - Feb 2023

University of Potsdam, Germany

Honolulu, HI (Remote)

- Successfully collaborated in a small teaching team to increase student satisfaction and maintain academic standards.
- Organized and graded weekly assignments for a cohort of 60+ students, ensuring timely distribution and clarity.
- Authored weekly comprehensive weekly sample solutions for machine learning problems, facilitating better understanding for students and the teaching team.
- Provided prompt, thorough responses to student inquiries on lecture material and homework, fostering engagement and comprehension.

Data Entry / Working Student in Radio Accreditation

Jul 2019 - Apr 2020

Continental Automotive GmbH

Regensburg, Germany

- Maintained and extended the database and documentation.
- Researched regulations and corresponded with worldwide public authorities.
- Processed and extracted data from certificates.
- Adjusted renewal process timeline of accreditation certificates to ensure continual product validity in one of their biggest markets.

PROJECTS

Skype Waddle (2023). Designed, developed, and deployed web app for processing and visualizing video chat history data. Utilized Pandas, Numpy, PlotlyDash, and Heroku. <https://github.com/stephanzwicknagl/skypewaddle>

ACTIVITIES

Board Member of East West Center Participants Association

2023 - 2024